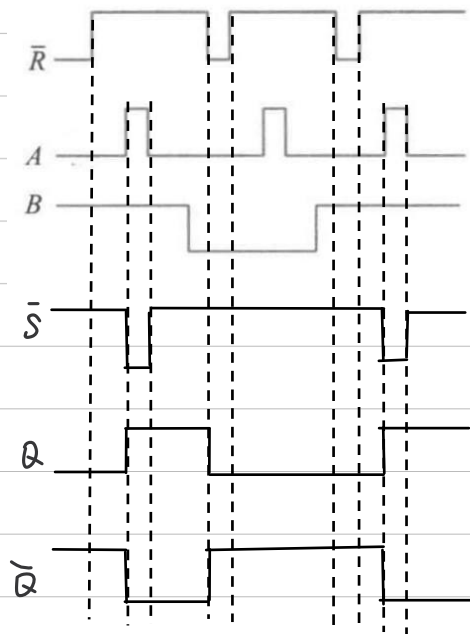


4.5.1

图中是一个 SR 触发器. 真值表如下:

$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$
0	1	1	0
1	0	0	1
1	1	保持	
0	0	禁止	

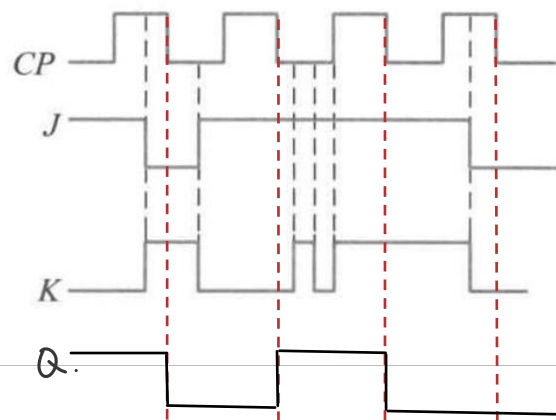
$$\bar{S} = \bar{A}B$$



4.5.4

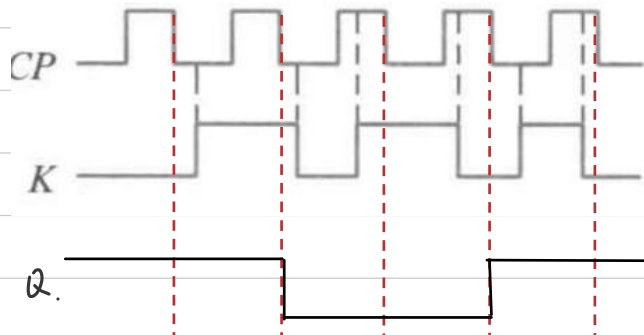
JK 触发器的真值表.

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$



4.5.5 由图得. CP 是负边沿触发.

$$J = \bar{K}$$



4.5.7

$$Q_{n+1} = D = \overline{Q_n T} \cdot T \overline{Q_n} = Q_n \overline{T} + T \overline{Q_n}$$

转换表:

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

4.6.1

输出方程: $Z = A Q_0 Q_1$

驱动方程: 对 FF₀ $J=A$ $K=A$

对 FF₁ $J=A Q_0$ $K=\overline{A Q_0}$

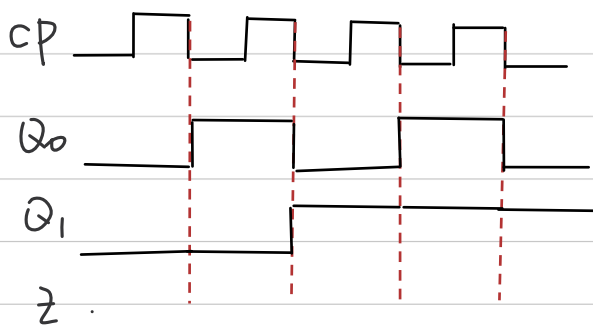
状态方程: 对 JK 触发器 $Q_{n+1} = J \overline{Q_n} + \overline{K} Q_n$

对 FF₀ $Q_0^{n+1} = A \overline{Q_0^n} + \overline{A} Q_0^n$

对 FF₁ $Q_1^{n+1} = A Q_0^n \overline{Q_1^n} + \overline{A Q_0^n} Q_1^n$

$A=1$. 假设初始条件为

(2)	Q_0^n	Q_1^n	Q_0^{n+1}	Q_1^{n+1}	Z
	1	1	0	0	0
	0	1	1	1	1
	1	0	0	1	0
	0	0	1	0	0



(3) 当 $A=0$ 时.

$$Q_0^{n+1} = Q_0^n$$

$$Q_1^{n+1} = Q_1^n$$

$$J_1 = J_2 = K_1 = K_2 = 0$$

$$Z = 0$$

即触发器的 J, K 输入均为 0.

输出 Z 恒为 0.

Q_0, Q_1 保持恒定.

4.6.2.

驱动方程:

对 FF_0 来说: $D_0 = \overline{Q_1^n} \overline{Q_0^n}$

对 FF_1 来说: $D_1 = Q_0^n$

状态方程:

$$Q_1^{n+1} = Q_0^n$$

$$Q_0^{n+1} = D_0 = \overline{Q_1^n} \overline{Q_0^n}$$

为什么不是 $Q_1^{n+1} = Q_0^{n+1}$?

状态转换表:

Q_0^n	Q_1^n	Q_0^{n+1}	Q_1^{n+1}
0	0	1	0
0	1	0	0
1	0	0	1

这里不会有 "11" *

波形图:

